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GEOG 482
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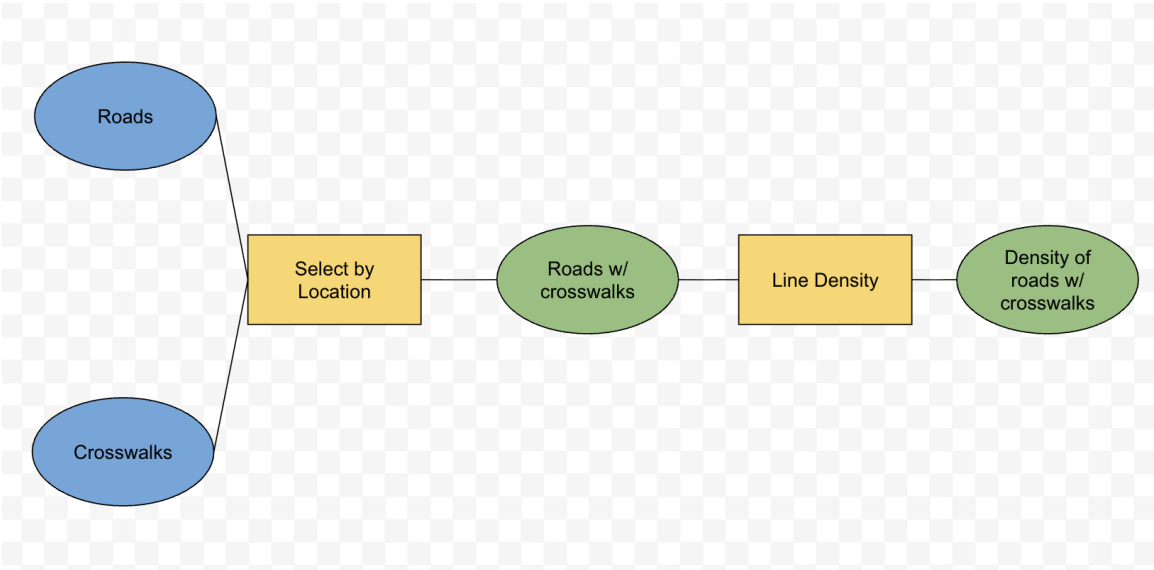
Lab 4 Write-up and Analysis Models

Sub-model 1: Accessibility for Pedestrians

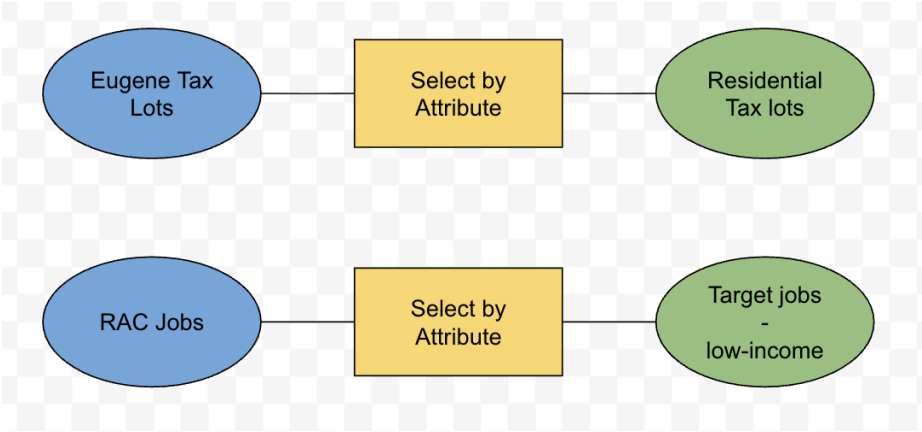
For my first model I wanted to find what areas would be the most accessible for pedestrians to get to the scooter hubs. The data I used were the roads dataset and the crosswalk dataset. I only wanted to focus on streets that have a crosswalk so, after I clipped both the data layers to my study area, I performed a 'Select by Location' of the roads that intersected with crosswalks. I wanted to show the areas with the most roads that had crosswalks so I used the Line Density tool to show this.



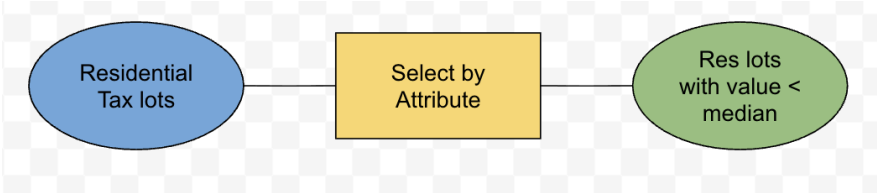
Sub-model 2: Demand by Commuters

To find the demand of who would be using these scooters the most I used the tax lots dataset, the RAC data from the census LODES dataset, and the LTD_busstops dataset. The RAC dataset classifies jobs by a number of features, such as worker ages, job earnings, and other demographic features. I wanted to focus on what I call "potential low-income workers", which include not only the obvious (jobs with low-income) but also jobs with young workers, and jobs with less education. The specific dataset I used was from the year 2017. I used 'Select by

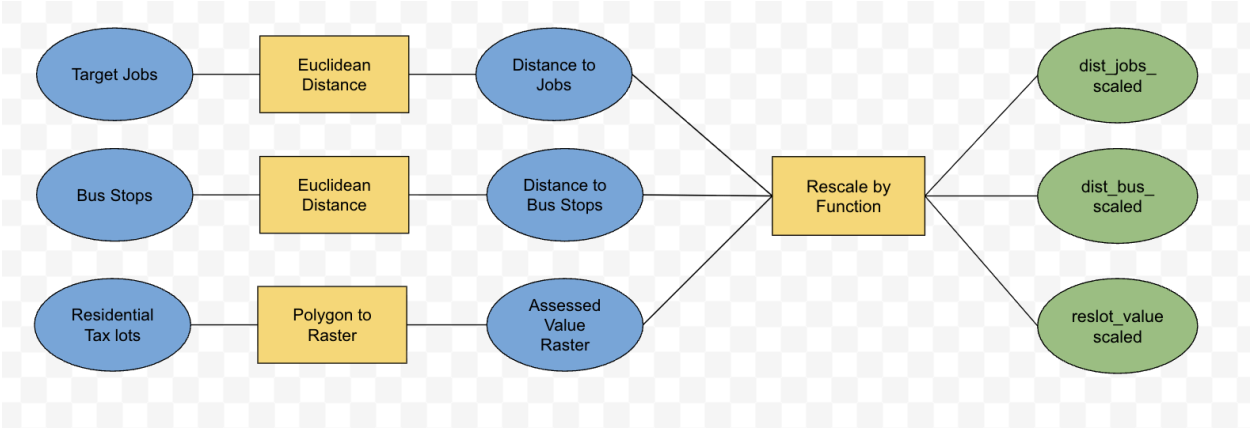
Attribute' to filter out the jobs within the parameters I mentioned. I was able to use the LODES documentation to find which fields to focus on.



As for processing the tax lot dataset, I simply selected the instances that were classified as both 'residential' and NOT 'single-family', because I wanted to aim at lower-income housing which would typically be apartments or multi-family housing. I then further selected out the lots with an assessed total value below the median.

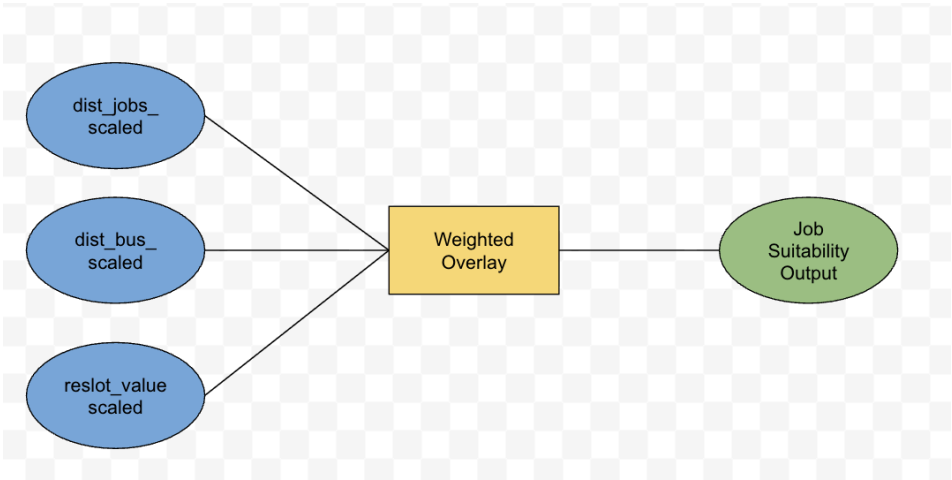


My goal was to create a weighted overlay that would score every residential tax lot by how likely it is to house low-income workers, combining distance, affordability, and demographics. I first calculated the Euclidean distance for each of my target jobs, and for the distance to each bus stop. I then needed to rasterize my tax lots, which I did based on the assessed value field. Then I needed to rescale all my raster layers to a common 1-10 scale.



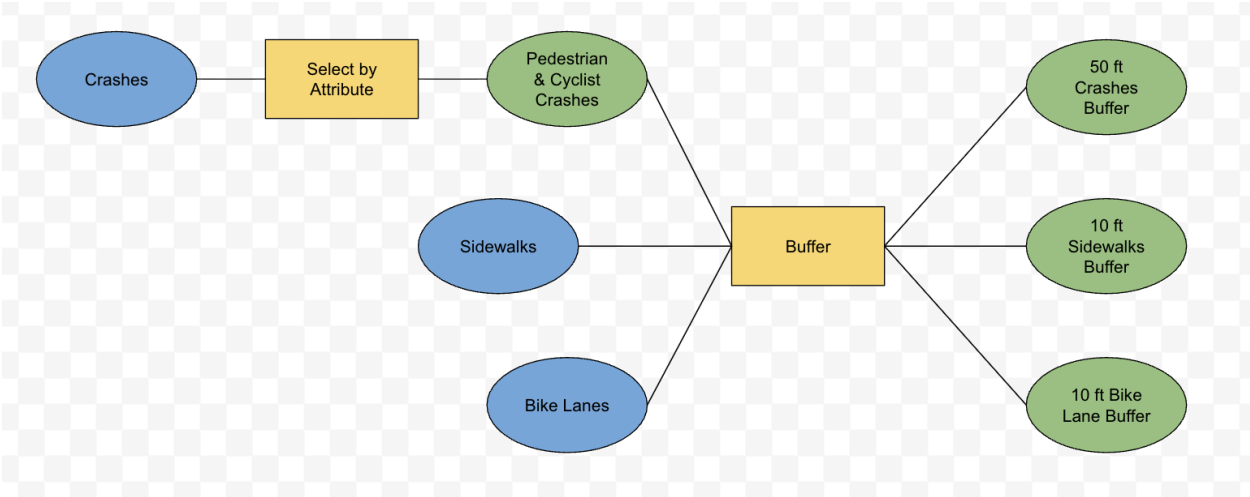
I input them each into the 'Weighted Overlay' tool, choosing the weights as shown in the table below.

Distance to Jobs	50%
Residence Lot Value	40%
Distance to Bus	10%

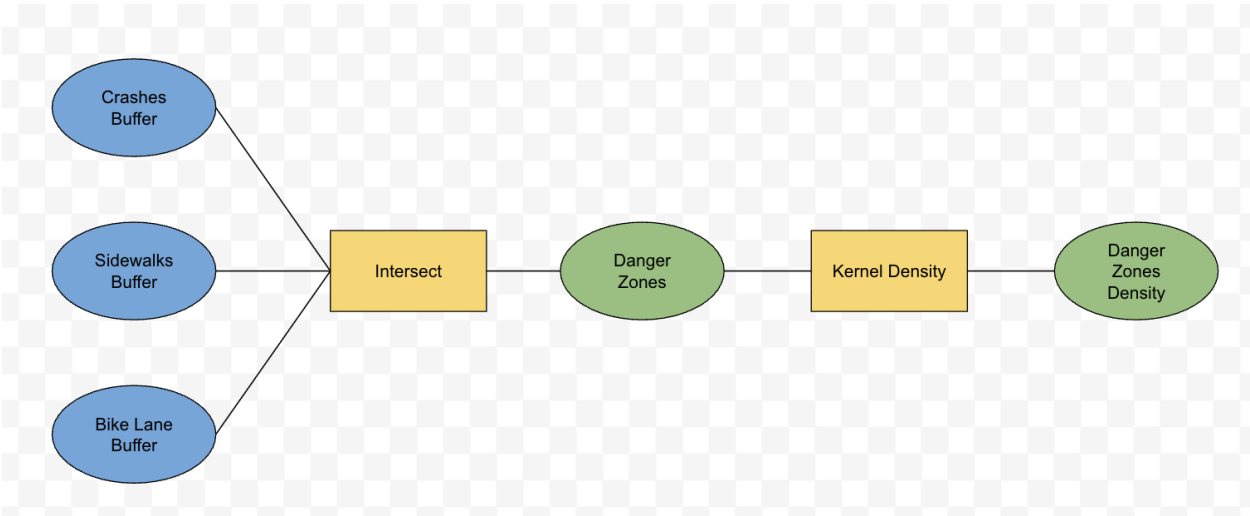


<u>Sub-model 3</u><u>: Safety for Riders</u>

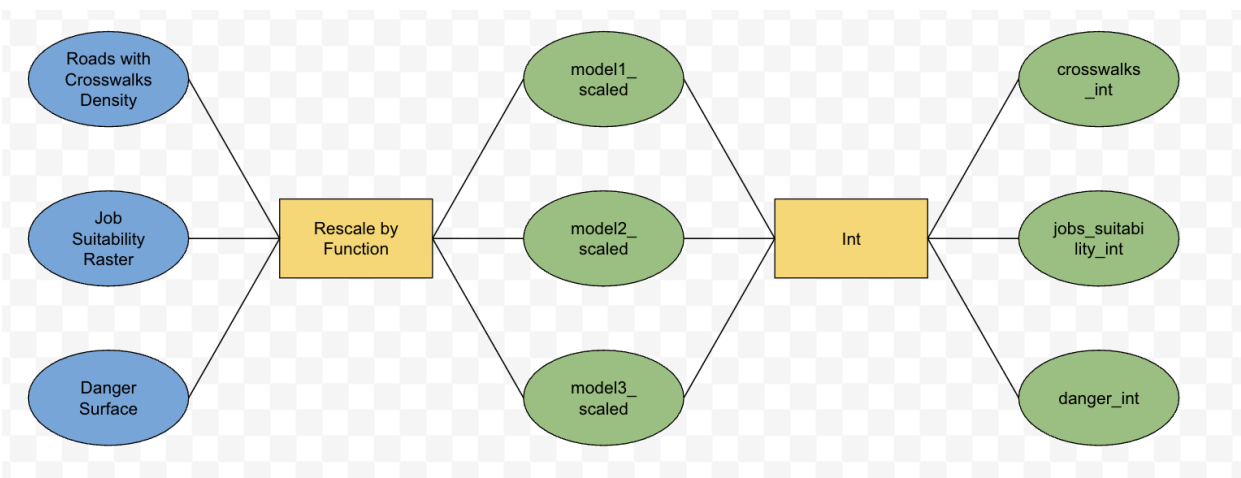
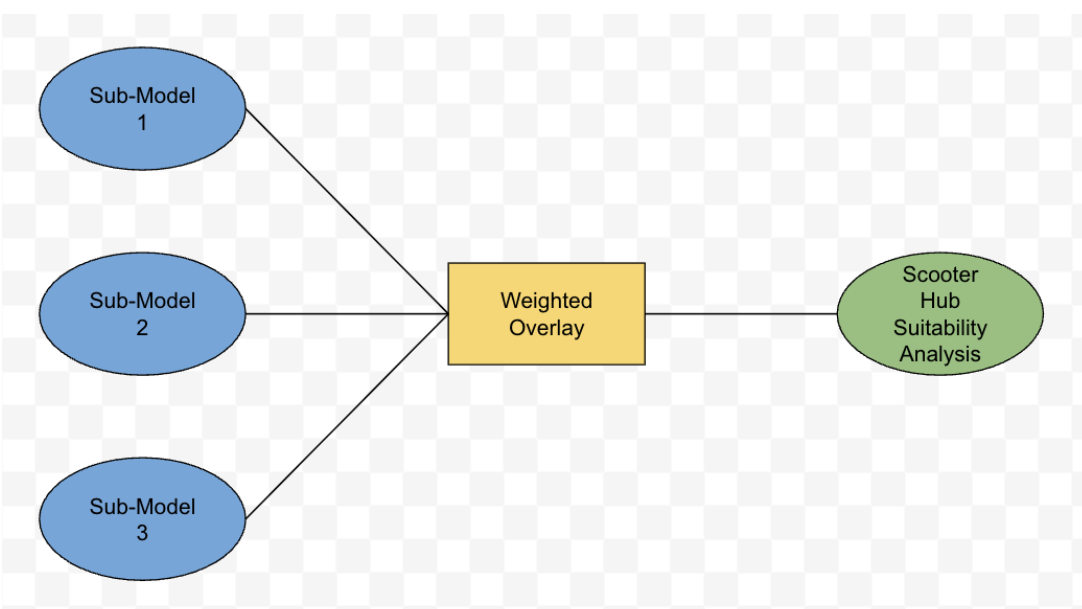
For my third and final sub model before completing my final analysis, I wanted to focus on safety for the scooters themselves. The safest places for scooters to ride is either on the sidewalk or in bike lanes, so I used polyline data from datasets for each of these. I also wanted to include areas with frequent vehicle crashes as less suitable, so I also used the crashes dataset. First, I selected from the crash dataset, only the incidents that involved pedestrians or cyclists. I then created a 50 ft buffer around the crash sites, and a 10 ft buffer around both the bike lanes and sidewalks. The buffer distance here matters because it defines how close something needs to be to a crash to be considered overlapping. I then used the 'Intersect' tool to find the intersection of all three of these things. These intersections represent the most dangerous places for scooter riders, thus I would want to avoid putting hubs near them.



Lastly, to create a raster layer I can use to contribute to my final map, I used the 'Kernel Density' tool to create a layer where higher values = more crashes concentrated in that area.



Putting it all together

Now it's time to put these three sub-models together to complete our analysis! First I needed to rescale each of my layers to normalize them, as well as convert them to integer values so I can perform a weighted overlay. Using the weights in the table below.

Pedestrian Accessibility	40%
Commuter Demand	35%
Rider Safety	25%